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SYSTEMATIC REVIEW

Just How Confident Can We Be in Predicting Sports Injuries? A Systematic Review of the Methodological Conduct and Performance of Existing Musculoskeletal Injury Prediction Models in Sport

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Abstract

Background An increasing number of musculoskeletal injury prediction models are being developed and implemented in sports medicine. Prediction model quality needs to be evaluated so clinicians can be informed of their potential usefulness.
Objective To evaluate the methodological conduct and completeness of reporting of musculoskeletal injury prediction models in sport.
Methods A systematic review was performed from inception to June 2021. Studies were included if they: (1) predicted sport injury; (2) used regression, machine learning, or deep learning models; (3) were written in English; (4) were peer reviewed.
Results Thirty studies (204 models) were included; 60% of studies utilized only regression methods, 13% only machine learning, and 27% both regression and machine learning approaches. All studies developed a prediction model and no studies externally validated a prediction model. Two percent of models (7% of studies) were low risk of bias and 98% of models (93% of studies) were high or unclear risk of bias. Three studies (10%) performed an a priori sample size calculation; 14 (47%) performed internal validation. Nineteen studies (63%) reported discrimination and two (7%) reported calibration. Four studies (13%) reported model equations for statistical predictions and no machine learning studies reported code or hyperparameters.
Conclusion Existing sport musculoskeletal injury prediction models were poorly developed and have a high risk of bias. No models could be recommended for use in practice. The majority of models were developed with small sample sizes, had inadequate assessment of model performance, and were poorly reported. To create clinically useful sports musculoskeletal injury prediction models, considerable improvements in methodology and reporting are urgently required.

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Key Points

Ninety-eight percent of sport musculoskeletal injury prediction models (and 79% of studies) were rated as high risk of bias.

Current musculoskeletal injury prediction models in sport are poorly developed and evaluated not applicable for further model validation by independent researchers due to high risk of bias and lack of reported equations and/or code.

To create clinically useful sports musculoskeletal injury prediction models, improving methodological conduct and transparent reporting is a priority.

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1 Introduction

Participating in sport carries inherent risk, resulting in musculoskeletal injuries [1] with repercussions of increased healthcare service burden [2], persistent joint pain [3, 4], post-traumatic osteoarthritis [5], and decreased quality of life [6]. Sports medicine clinicians perform periodic health examinations [7] to inform the prescription of universal, selective, or individualized interventions that are designed to mitigate injury risk [8]. However, while these types of management decisions are reliant upon clinical knowledge, skills and intuition [9], it is not always clear who is at risk or will sustain an injury [10]. As a result, sport organizations and clinicians implement heuristics (known as 'rules of thumb') [11] and clinical prediction models [12] to provide a more objective insight into an athletes' injury risk [9, 13].

Clinical prediction models were first used through the Framingham Heart Study [14], but have diversified into fracture [15] and critical care [16]. Within sports medicine, the development of clinical prediction models has been low compared to those developed in general medical fields, but this is now increasing [17, 18]. In particular, clinical prediction models use multiple variables (also known as predictors) to calculate the risk or probability of a health outcome (such as musculoskeletal injury) for an individual [19]. Prediction models have traditionally been developed through statistical approaches such as regression, but more increasingly they are being developed using machine learning methods [18, 20]. Machine learning methods generally impose minimal modelling assumptions [21], potentially offering increased flexibility to analyze higher order interactions and non-linear associations [22], but at the expense of requiring larger sample sizes [23]. However, in terms of predictive accuracy, there is often no discernible difference between the two broad classes of approaches [24]. Regardless of the modelling method, poor methodological conduct and reporting leading to high risk of bias has been observed in studies in other healthcare fields [24, 25].

Clinical prediction model quality and transparency are essential to have any chance of further evaluation and implementation in to patient care [9, 13, 26]. Due to the importance of this issue, the Transparent Reporting of Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) statement was developed in 2015 [26]. As prediction models may perform differently in diverse athlete or sport settings [27], depending on the intended athlete population, models need to be evaluated in individuals from different geographical regions or competition levels [27, 28]. Models should also undergo regular evaluation to maintain predictive accuracy in contemporary populations. Further, the full model must be reported (model availability or code provided in the context of machine learning) to allow independent researchers to evaluate and understand if a prediction model is clinically useful in their target population, and can be implemented clinically. While sports musculoskeletal injury prediction models have been previously evaluated through systematic reviews [17, 18, 29], these reviews were limited in their search comprehensiveness, data extraction, and evaluation of methodology, reducing a clinician's ability to fully critically appraise these models and their potential usefulness within their clinical setting [30]. Therefore, existing models may not be fit for purpose, potentially even harming patient care, and often lack the transparency to facilitate evaluation by independent research groups. To address this, this article provides a systematic review to evaluate the methodological conduct and completeness of reporting of musculoskeletal injury prediction models in sport. Our aim is to provide sports medicine clinicians and scientists with a critical evaluation of the current state of sports musculoskeletal injury prediction models, to better inform their conduct and clinical usefulness.

2 Methods

2.1 Study Design

The Critical Appraisal and Data Extraction for Systematic Reviews of Prediction Modelling Studies Checklist (CHARMS) [31] was used to frame the systematic review and develop the data extraction protocol. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline was used to guide the reporting of this systematic review. Please refer to Appendix 1 in the Online Supplemental Material (OSM) for framing of the review through the CHARMS Checklist [31]. This review was prospectively registered on the Open Science Framework (OSF) and is available at: https://osf.io/52mzn/.

2.2 Inclusion and Exclusion Criteria

Studies were included if they developed or evaluated a model that predicted injury in sport. Injury prediction had to refer to predicting the occurrence, the severity, or the type of musculoskeletal injuries [17]. A musculoskeletal injury was defined as an injury to a tendon, ligament, nerve, muscle or bone that occurs during any sport training, matches, competition or event.

Inclusion criteria consisted of:

- A description of the development or validation of a clinical prediction model for individualized musculoskeletal

injury prediction using two or more predictors in a sport setting.

- Prediction models developed using regression (e.g., logistic or cox regression), machine learning (e.g., tree-based methods, support vector machines, ensemble methods), or deep learning (e.g., artificial neural networks).
- English language only.
- Peer-reviewed articles (i.e., preprints not included).

Exclusion criteria consisted of:

- Studies that investigated associations of individual factors.
- A new modeling approach was introduced (hence a methodological focus).
- Reviews of the literature.
- Studies for which we were unable to obtain the full text.
- Studies that predict concussion only.

2.3 Search Strategy

Databases: Medline (Ovid), EMBASE, CENTRAL, SPORTDiscus and CINAHL (Cumulative Index for Allied Health Literature) electronic databases were searched from inception to 24 June 2021 to identify relevant articles. All searches were documented including terms used and number of hits or articles obtained. Bibliographies of selected studies were hand searched to identify relevant articles not found by the search strategies. Medical subject headings (MeSH) and keywords were utilized, including: "ROC Curve", "risk scores", "predict", "logistic", "regression", "injury", and "sport" (Appendix 2, OSM).

2.4 Study Selection

Title and abstracts were first screened by two blinded reviewers (GB, JM) using inclusion and exclusion criteria. Two blinded reviewers (GB, JM) performed a full-text review following title and abstract screening. Any conflicts were first discussed between both reviewers. If a consensus could not be reached, another reviewer (TH) was utilized to determine final study eligibility. Following full-text review, a hand search was performed for any studies missed within the initial search (Fig. 1).

2.5 Data Extraction

Data were extracted into a customized Excel spreadsheet (Version 2013, Microsoft, Redmond, WA, USA) by two independent reviewers (GB, JM) (Table 1). Disagreements concerning data domain placement were resolved by a third reviewer (TH). Raw data were uploaded to the Open Science Framework.

2.6 Risk of Bias

Risk of bias is defined as shortcomings in the study design, conduct and analysis that can result in the findings being flawed or distorted [32]. We assessed risk of bias for each model (and also summarized at the study level) with the Prediction Model Risk of Bias Assessment Tool (PROBAST) [32]. PROBAST identifies sources of bias in four domains: participants, predictors, outcomes and analysis. A PROBAST assessment was carried out independently by two reviewers (GB, JM). A third reviewer resolved any conflicts (TH).

2.7 Statistical Analyses

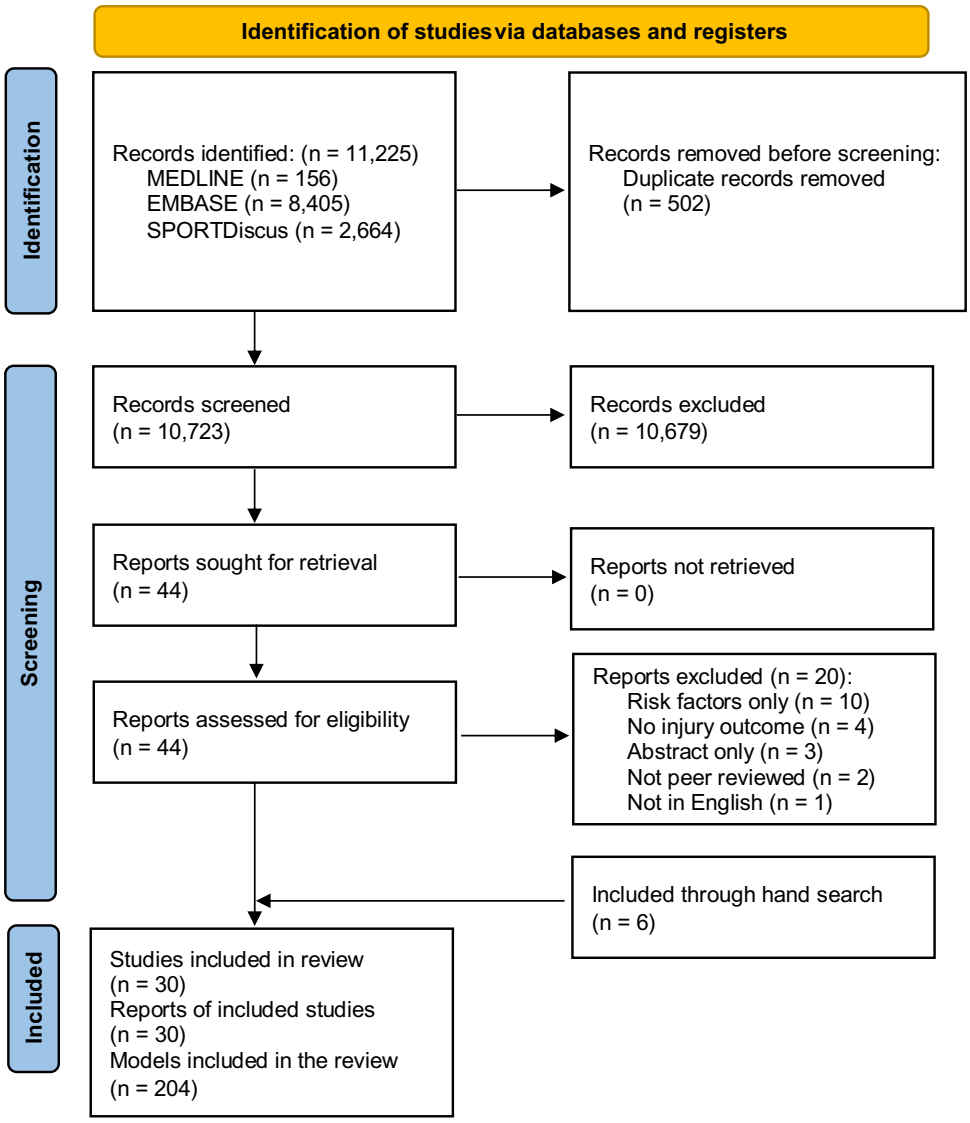
Data were summarized using counts (percentages) and medians (ranges). Depending on whether the extracted item is fixed at the study level or varied at the model level (depending on the modelling approach), data were summarized accordingly. All analyses were performed in R version 4.02 (R Core Team (2013).

3 Results

3.1 General Summary

A total of 10,723 titles and abstracts were screened, with 30 studies (reporting the development of 204 models) included (Fig. 1) [33–62]. The 30 studies collectively included 9,142 participants, with 6,549 males (median: 96; range 25–2322), and 860 females (median: 105; range 26–187). Ten (33%) studies included soccer athletes [34, 39, 44, 45, 47, 48, 50–52, 62], five (17%) athletes from undefined sports [33, 42, 43, 49, 55], five (17%) basketball athletes [41, 44, 54, 56, 58], two (7%) Australian Rules Football athletes [36, 53], two (7%) volleyball athletes [44, 58], and 12 (40%) studies included athletes from baseball, floorball, football, gymnastics, handball, korfball, ice hockey, rugby, or swimming [35, 37, 38, 41, 45, 46, 49, 57–61]. Fourteen (47%) studies included professional or elite athletes [33, 34, 36, 38, 39, 45, 46, 52–54, 56–59, 62], eight (27%) college athletes [35, 44, 47, 49, 50, 55, 57, 60, 61], five (17%) youth athletes [37, 40, 41, 48, 51], two (7%) recreational adult athletes [43, 54], and two (7%) high school athletes [42, 55]. Twenty-one (70%) studies used a prospective cohort design [33, 34, 37, 38, 40–42, 44, 45, 47, 48, 50–54, 57, 58, 60, 61], five (17%) studies were retrospective cohort studies [35, 36, 39, 46, 56], and four (13%) studies used a case control study design [43, 49, 55, 59]. Eighteen (60%) studies developed models using regression [33, 35–40, 43, 44, 47, 49, 50, 54–56, 58, 60, 61], four (13%) studies used machine learning methods [34, 45, 51, 52], seven (23%) studies investigated both regression and

Fig. 1 PRISMA flow diagram



machine learning methods [41, 42, 46, 48, 57, 59, 62], and one study (3%) investigated regression, machine learning, and deep learning models [53]. All 30 studies focused on model development [33–62], and no studies carried out an external validation or model updating (Table 2).

3.2 Risk of Bias

For risk of bias for each study, please refer to Appendix 3 (OSM). Two (7%) studies were rated overall as low risk of bias [37, 39] and 28 (93%) studies were rated as high or unclear risk of bias (Table 3) [33–36, 38, 40–62]. For the participants domain, 26 (87%) studies [34–48, 50–54, 56–58, 60–62] were rated as low risk of bias and four (14%) studies were rated high risk of bias [33, 49, 55, 59]. For the predictor domain, 23 (77%) studies were rated low risk of bias [33, 34, 36–48, 50–54, 56, 57, 61] and seven (23%) studies were rated as unclear risk of bias [35, 49, 55, 58–60, 62]. For the outcome domain, 23 (77%) studies were rated as low risk of bias [34, 36–45, 47, 48, 50–54, 56, 57, 60–62], six (20%) studies were rated as unclear risk of bias [33, 35, 49, 55, 58, 59], and one (3%) study was rated as high risk of bias [46]. For the analysis domain, two (7%) studies were rated as low risk of bias [37, 39] and 28 (93%) studies were rated as high or unclear risk of bias [33–36, 38, 40–62].

3.3 Sample Size and Predictors

The median sample size used to develop the musculoskeletal injury prediction models was 152 (range 25–2322), and the median number of injuries (outcome to be predicted) was 57 (range 5–6673). The median number of events per parameter was 2.7 (range 0.3–78.6). Three studies (10%) [37, 39, 40] performed an a priori sample size calculation, and one (3%) of these [39] performed a sample size calculation using the newly developed guidance [63]. Only one regression model explored whether

Table 1 Variables extracted, according to critical appraisal and data extraction for systematic reviews of prediction modelling studies: the CHARMS Checklist [27]

Domain	Key items
Source of data	Study design and source
Participants	Eligibility; recruitment; participant characteristics; sport, gender, country, level of sport
Outcome to be predicted	Type of musculoskeletal injury, body part(s), recurrence, follow up time of the outcome
Candidate predictors	Number and type of predictors; timing of predictors; whether or not predictors were measured and examiners were blinded to the outcome; handling method of any continuous predictors
Sample size	Details of any sample size calculation. The overall sample size and number of outcome events.
Missing data	Number of missing data; handling of missing data
Model development	Modeling method (e.g., logistic regression, random forest, neural network); approach for any predictor selection before model development; method for predictor selection during model development; whether shrinkage performed (for regression models) and if so how; Machine learning tuning hyperparameters
Model performance	Calibration and how it was reported (e.g., calibration slope, graphical calibration plot), discrimination, net benefit, other performance measures
Model evaluation	Whether internal validation was performed, if so, what type (e.g., bootstrap, cross-validation, split-sample); details of any external validation performed In case of poor validation, whether model was adjusted or updated (e.g., intercept recalibration, predictor adjustment for regression models)
Results	Whether the final model equation was reported, with intercept and predictor coefficients (for regression models) Whether the code was made available for machine learning models, and were all hyperparameters reported? Whether alternative methods were used to present the model and if so, what types (e.g., nomogram)? Whether a comparison of performance was performed, and distribution of predictors with and without missing data or handling of missing data (e.g., imputation) Whether study was reported according to TRIPOD Guidelines [79]

TRIPOD Transparent Reporting of Multivariable Prediction Model for Individual Prognosis or Diagnosis

continuous variables needed to be modelled to account for any non-linearity [39], and others did not justify why they did not. Ten studies (33%) dichotomized continuous predictors prior to developing the model [36, 42–45, 47–49, 60, 61], and an additional three studies (10%) were unclear about whether continuous predictors were dichotomized prior to developing the model [33, 34, 50]. Twelve studies (40%) selected predictors through univariable analyses (i.e., based on their unadjusted association with the outcome) before model development [36, 37, 43, 48, 49, 54, 56, 57, 59, 61], seven studies (23%) were unclear on how predictors were selected before or during model development [38, 40, 45, 46, 50, 58], one study (3%) used recursive cross-validation methods (sic) [52], four (13%) included all predictors [44, 47, 51, 53], three (10%) used backwards selection [33, 35, 39], three (10%) used least absolute shrinkage and selection operator (LASSO) [41, 42, 62], and two (7%) included both univariable (before model development) and backward step-wise selection (during model development) [55, 60]. Three machine learning studies (10%) [45, 52, 53] over-sampled to reportedly control for class imbalance, and two (7%) [45, 62] under-sampled to control for class imbalance.

3.4 Missing Data

Six (20%) studies reported missing data [37, 39, 41, 42, 45, 62] and 24 (83%) studies did not mention missing data [33–36, 38, 40, 43, 44, 46–61]. Of the six (20%) studies that reported missing data [37, 39, 41, 42, 45], two (66%) performed multiple imputation [37, 39], one (3%) performed k-nearest neighbors imputation [41], two (7%) imputed the mean for each predictor [45], and one (3%) did not report having any missing data; thus a complete case analysis was performed [42]. Both studies that performed multiple imputation also imputed the outcome [37, 39].

3.5 Algorithms Implemented

In the 18 (62%) studies that developed models only using regression [33, 35–40, 43, 44, 47, 49, 50, 54–56, 58, 60, 61], 15 (75%) models were developed with logistic regression [33, 35, 37–39, 43, 44, 47, 49, 50, 54, 55, 58, 60, 61], two (10%) models were developed through Poisson regression [36, 56], one (5%) model was developed with a latent classification model [40], and one (5%) model was developed with Cox regression [61]. In the four (14%) studies that only used machine learning methods [34, 45,

Table 2 Description of included studies

Study	Year	Study design	Sample size	Number of events	Sport	Competition level	Country	Injured body part(s)	Total number of models	Prediction methods
Abalo-Nunez et al. [33]	2018	Prospective cohort	73	12	Gymnastic, "other"	Elite	Spain	Lower extremity	1	Logistic regression
Ayala et al. [34]	2019	Prospective cohort	96	18	Soccer	Professional	Spain	Hamstring	18	Tree Ensemble
Carbuhn et al. [35]	2020	Retrospective cohort	27	5	Cross-country	College	USA	Long bone	1	Logistic regression
Carey et al. [62]	2018	Prospective cohort	75	156	Soccer	Professional	Australia	Hamstring	4	Logistic regression, Generalized estimating equations, Tree ensemble, Support vector machine
Colby et al. [36]	2018	Retrospective cohort	60	58	Australian Rules Football	Professional	Australia	Lower extremity	2	Poisson regression
Feijen et al. [37]	2021	Prospective cohort	201	42	Swimming	Youth	USA	Shoulder	1	Logistic regression
Gabbet [38]	2010	Prospective cohort	91	159	Rugby	Professional	Australia	Trunk, core, lower extremity	1	Logistic regression
Hughes et al. [39]	2020	Retrospective cohort	317	138	Soccer	Professional	UK	Lower extremity	1	Logistic regression
Ivarsson et al. [40]	2014	Prospective cohort	101	67	Soccer	Elite Junior	Sweden	Any MSK injury	1	Latent classification
Jauhiainen et al. [41]	2020	Prospective cohort	314	57	Basketball, floorball	Youth	Finland	Ankle, knee	2	Logistic regression, tree ensemble
Karuc et al. [42]	2021	Prospective cohort	556	90	Athletic, Non-Athletic	Adolescents	Croatia	Unclear	4	Logistic regression, Naïve Bayes, Linear discriminant, k-nearest neighbors
Khayambashi et al. [43]	2015	Case control	501	15	Athletes	Recreational	Iran	ACL	1	Logistic regression
Landis et al. [44]	2018	Prospective cohort	187	17	Soccer, volleyball, basketball	College	USA	Lower extremity	1	Logistic regression
Lopez-Valencia et al. [45]	2018	Prospective cohort	132	32	Soccer, handball	Professional	Spain	Lower extremity	68	Tree ensemble
Luu et al. [46]	2020	Retrospective cohort	2322	6673	Ice Hockey	Professional	USA	Any MSK injury	12	Logistic regression, k-nearest neighbors, Naïve Bayes, Boosting
McCann et al. [47]	2018	Prospective cohort	43	8	Soccer	College	USA	Ankle sprain	1	Logistic regression
Oliver et al. [48]	2020	Prospective cohort	335	99	Soccer	Youth	UK	Lower extremity	57	Logistic regression, Tree ensemble
Pontillo et al. [49]	2021	Case control	278	16	Athletes	College	USA	ACL	1	Logistic regression

Table 2 (continued)

Study	Year	Study design	Sample size	Number of events	Sport	Competition level	Country	Injured body part(s)	Total number of models	Prediction methods
Powers et al. [50]	2018	Prospective cohort	43	8	Soccer	College	USA	Ankle	1	Logistic regression
Rommers et al. [51]	2020	Prospective cohort	734	368	Soccer	Youth	Belgium	Any MSK injury	9	Gradient boosting machine
Rossi et al. [52]	2018	Prospective cohort	26	23	Soccer	Professional	Italy	Lower extremity	5	Tree ensemble
Ruddy et al. [53]	2018	Prospective cohort	362	53	Australian Rules Football	Professional	Australia	Hamstring	5	Logistic regression, Naïve Bayes, Tree ensemble, Support vector machine, Neural network
Shambaugh et al. [54]	1991	Prospective cohort	45	15	Basketball	Recreational	USA	Foot, ankle, knee, hip	1	Logistic regression
Sturnick et al. [55]	2015	Case control	176	88	Athletes	High School; College	USA	Knee	1	Logistic regression
Teramoto et al. [56]	2017	Retrospective cohort	1443	681	Basketball	Professional	USA	Any MSK injury	1	Poisson regression
Thornton et al. [57]	2017	Prospective cohort	25	156	Rugby	Professional	Australia	Any MSK injury	2	Generalized Estimating equations, Tree ensemble
van der Does et al. [58]	2015	Prospective cohort	75	11	Basketball, volleyball, korfball	Elite or Sub-Elite	Netherlands	Ankle and knee	1	Logistic regression
Whiteside et al. [59]	2016	Case control	208	104	Baseball	Professional	USA	UCL	3	Logistic regression, Naïve Bayes, Support vector machine
Wiese et al. [60]	2014	Prospective cohort	144	93	Football	College	USA	Upper or lower extremity	1	Logistic regression
Wilkerson et al. [61]	2015	Prospective cohort	152	132	Football	College	USA	Core or lower extremity	2	Logistic regression, Cox regression

ACL anterior cruciate ligament, *MSK* musculoskeletal, *UCL* ulnar collateral ligament

Table 3 Risk of bias using Prediction Model Risk of Bias Assessment Tool (PROBAST)

Study	Overall	Analyses	Outcome	Predictors	Participants
Abalo-Nunez et al. [33]	High	High	Unclear	Low	High
Ayala et al. [34]	High	High	Low	Low	Low
Carbuhn et al. [35]	High	High	Unclear	Unclear	Low
Carey et al. [62]	High	High	Low	Unclear	Low
Colby et al. [36]	High	High	Low	Low	Low
Feijen et al. [37]	Low	Low	Low	Low	Low
Gabbet [38]	High	High	Low	Low	Low
Hughes et al. [39]	Low	Low	Low	Low	Low
Ivarsson et al. [40]	High	High	Low	Low	Low
Jauhiainen et al. [41]	Unclear	Unclear	Low	Low	Low
Karuc et al. [42]	High	High	Low	Low	Low
Khayambashi et al. [43]	High	High	Low	Low	Low
Landis et al. [44]	High	High	Low	Low	Low
Lopez-Valencia et al. [45]	High	High	Low	Low	Low
Luu et al. [46]	Unclear	Unclear	High	Low	Low
McCann et al. [47]	High	High	Low	Low	Low
Oliver et al. [48]	High	High	Low	Low	Low
Pontillo et al. [49]	High	High	Unclear	Unclear	High
Powers et al. [50]	High	High	Low	Low	Low
Rommers et al. [51]	High	High	Low	Low	Low
Rossi et al. [52]	High	High	Low	Low	Low
Ruddy et al. [53]	Unclear	Unclear	Low	Low	Low
Shambaugh et al. [54]	Unclear	Unclear	Low	Low	Low
Sturnick et al. [55]	High	High	Unclear	Unclear	High
Teramoto et al. [56]	High	High	Low	Low	Low
Thornton et al. [57]	High	High	Low	Low	Low
van der Does et al. [58]	High	High	Unclear	Unclear	Low
Whiteside et al. [59]	High	High	Unclear	Unclear	High
Wiese et al. [60]	High	High	Low	Unclear	Low
Wilkerson et al. [61]	High	High	Low	Low	Low

Risk of bias was assessed at the model level. Risk of bias is reported for all models at the study level unless differences in model risk of bias within specific studies were reported

51, 52], 27 (28%) models were developed using random forest or decision trees [34, 52], one (1%) model was developed using gradient boosting machines [51], and 68 (71%) models were unclear on the proportion of different machine learning methods employed [45]. In the eight (27%) studies that implemented both regression and machine learning based approaches [41, 42, 46, 48, 53, 57, 59, 62], eight (10%) models were logistic regression [41, 42, 46, 48, 53, 59, 62] and two (3%) models were a generalized estimating equation [57, 62]. Concerning the machine and deep learning models, 63 (79%) models were random forest or decision trees [41, 48, 53, 57, 59, 62], four (5%) models were Naïve Bayes [42, 46, 53, 59], two (3%) models were k-nearest neighbors [42, 46], three (4%) models were support vector machines [53, 59, 62], two (3%) models were gradient boosting machines [46], one (1%) model was linear discriminant [42], and one (1%) model was neural network [53].

3.6 Validation

Fourteen studies (47%) performed internal validation [34, 36, 37, 39, 41, 42, 45, 46, 51–53, 57, 59, 62]. Two studies (7%) [37, 39] performed internal validation using bootstrapping, nine studies (30%) used cross-validation [34, 36, 41, 42, 45, 46, 53, 59, 62], and four studies (13%) used split-sample methods [51, 52, 57]. The median split-sample split was 70% (range 70–80) for model development, and 20% (range 15–30) for model evaluation. Two studies (7%) performed LASSO for shrinkage of coefficients [41, 42].

Table 4 Musculoskeletal injury prediction model performance statistics for each included study

Study	Discrimination	Accuracy	Calibration	R^2	Sensitivity	Specificity
Abalo-Nunez et al. [33]					0.58	0.97
Ayala et al. [34]	0.84				0.78	0.84
Carbuhn et al. [35]	0.87	0.96				
Carey et al. [62]	0.72					
Colby et al. [36]	0.54–0.59					
Feijen et al. [37]	0.71 (0.60–0.94)		Calibration Plot; Hosmer & Lemeshow Test	0.14		
Gabbet [38]		0.62			0.87 (0.81–0.92)	0.99 (0.98–0.99)
Hughes et al. [39]	0.59 (0.53–0.65)		0.72 (0.28–1.61)	0.06		
Ivarsson et al. [40]				$p < 0.001$		
Jauhiainen et al. [41]	0.65					
Karuc et al. [42]	Logistic regression: 0.56 (0.47–0.63) Naïve Bayes: 0.58				Logistic regression: 0.50 Naïve Bayes: 0.70	Logistic regression: 0.55 Naïve Bayes: 0.49
Khayambashi et al. [43]	0.78 (0.70–0.86)			0.11	0.93 (0.68–0.99)	0.87 (0.60–0.98)
Lopez-Valencia et al. [45]	0.75				0.66	0.79
Luu et al. [46]	Logistic regression: 0.94 XGBoost: 0.95	Logistic regression: 0.95 XGBoost: 0.95				
McCann et al. [47]	0.73 (0.58–0.89)				0.88	0.51
Oliver et al. [48]	0.66 (0.60–0.73)				0.74	0.56
Powers et al. [50]	0.67 (0.56–0.78)			0.07	0.48 (0.28–0.69)	0.82 (0.75–0.88)
Rommers et al. [51]		0.85		0.85	0.85	
Rossi et al. [52]	Decision tree: 0.76				Decision tree: 0.87	Decision tree: 0.96
Ruddy et al. [53]	0.53					
Thornton et al. [57]	0.64					
Whiteside et al. [59]		Support vector machine: 0.75			Support vector machine: 0.74	Support vector machine: 0.75
Wiese et al. [60]	Unclear discrimination performance					
Wilkerson et al. [61]			Hosmer & Lemeshow Test		0.56	0.80

95% confidence intervals are reported in parentheses where available. When multiple machine learning models were developed, the best performing model is reported. Discrimination is reported as area under the curve (AUC)

3.7 Performance

Twenty-four (80%) studies reported measures of musculoskeletal injury prediction model performance (Table 4) [33–43, 45–48, 50–53, 57, 59–62]. Six (21%) studies did not report any musculoskeletal injury prediction model performance measures [44, 49, 54–56, 58]. Nineteen (63%) studies reported discrimination [34–37, 39, 41–43, 45–48, 50, 52, 53, 57, 60–62], three (10%) studies reported calibration [37, 39, 61], two (6%) studies reported the Hosmer and Lemeshow test [37, 61], four (13%) studies reported R^2 [37, 39, 43, 50], one (2%) study reported clinical decisions curves and net benefit [39], seven (23%) studies reported overall accuracy [34, 35, 38, 51–53, 59], and one (2%) study reported root mean square error approximation [40].

3.8 Model Presentation

Four (14%) studies reported model equations for the regression based approaches [33, 35, 39, 54], and one (3%) study partially reported model equation, without the intercept [37]. No machine learning studies or models reported tuned hyperparameters within the article, in an appendix, or provided a web link to code or logic. Out of the 29 studies [33–53, 55–62] that published following the release of the TRIPOD statement, two (7%) studies cited or discussed the TRIPOD guidelines [39, 42]. One (3%) study was published prior to TRIPOD development [54].

4 Discussion

Only two studies were rated at low risk of bias, with 28 studies rated as high risk of bias. Following PROBAST, studies are considered to be at high risk of bias overall when one or more appraisal domains are judged to be at high risk [32]. When stratified according to PROBAST domains, the poor overall quality of the current evidence could be explained because, crucially, the analyses were considered at high risk of bias for all of these 28 studies.

While the performance metrics across these reported 28 studies varied, some reported high levels of discrimination [46], accuracy [35], sensitivity [38, 43, 47, 51], and specificity [33, 43, 48, 50], so it would be tempting to assume that some models may offer clinical value. Importantly for clinicians, however, our risk of bias assessment suggests that the reported performance metrics for these 28 studies are likely to be optimistic and considered as unreliable. Indeed, even if any of these existing models were to be deployed in practice with other athlete groups, it is likely that prediction performance would deteriorate, thus increasing the uncertainty regarding their clinical utility [64].

Of the two studies judged to be of low risk of bias [39, 42], their performance measures were modest, and it is unclear whether they would offer any clinical benefit, without further external validation and clinical implementation. Consequently, this means that there are no existing musculoskeletal injury prediction models that could be confidently recommended for use in sports medicine practice, and our review has clearly shown that a drive to improve research methodology (with particular emphasis on the conduct of analyses) is urgently required in this potentially important area.

The median sample size events per parameter for these musculoskeletal injury prediction models was 2.7 outcomes (i.e., injury) per predictor parameter. While prediction sample size calculations have moved beyond the 'events-per-variable' rule of thumb [65], this calculation still provides insight into the number of predictors in relation to the number of events (i.e., injury). Further, only three models included an a priori sample size calculation. Sample size guidance for developing clinical prediction models using regression have recently been proposed, and one model in the review followed these principles [63, 66]. There is no equivalent guidance for sample size of machine learning studies, so until guidance is made available, the aforementioned studies should be considered when calculating the required sample size [63, 66], bearing in mind these will likely be an underestimate as studies have shown machine learning methods require larger sample sizes compared to regression [23]. The small sample sizes and lack of sample size calculations suggest these models are at risk for overfitting. Overfitting is defined as when the model is too closely fitted to the idiosyncrasies in the data, decreasing the model's generalizability and usefulness in other samples [67]. Greater care is needed to perform a priori sample size calculations, to gain appropriate-sized datasets, and to increase prediction generalizability for future athlete injury risk assessment.

Almost half of the included studies dichotomized continuous predictors prior to incorporating into the model. This is where, prior to being incorporated into models, predictors derived from continuous measurements are converted into low- and high-risk groups using arbitrarily chosen or data-driven thresholds. Dichotomizing continuous predictors is discouraged as this method discards information, decreases prediction accuracy, and is biologically implausible [68]. Continuous predictors should remain continuous and be modelled appropriately (e.g., when developing a regression model to consider fractional polynomials or splines) [69]. Many machine learning methods (e.g., random forests) handle nonlinearity directly, but dichotomizing or categorizing prior to fitting the machine learning method should clearly be avoided. Linearity should also not be assumed as many biological relationships are non-linear in nature (e.g., body mass index and mortality risk) [68], and including non-linear transformations has demonstrated improved prediction in sport [70]. However, only one study [39] assessed for non-linear relationships between continuous predictors and the outcome. Further, almost one in two studies used univariable screening for predictor selection prior to any model building. While there may be more predictors available than can be included in the model, univariable selection methods should be avoided as certain predictors may only become significant for inclusion after controlling for other predictors or spurious univariable associations may cause other predictors to be included [71]. If predictor selection is required for regression-based models to control for overfitting, then approaches based on penalization such as LASSO or elastic net may be considered [72], though recent studies have shown these methods to be unreliable when the sample size is small [73].

Just under half of the studies performed internal validation, with one in four that performed internal validation using a split-sample approach. As prediction models are anticipated to have improved prediction performance during development compared to new data [74], it is important to control for this optimism bias through internal validation [75]. Internal validation can be performed through bootstrapping or cross-validation methods [13]. Split sampling methods have been advised against in the regression framework as being inefficient, as well as decreasing the development dataset size (thus increasing the risk of overfitting) [75], but are common in machine learning (where data are split into training, tuning and testing). As almost all of the included studies involved small sample sizes, split sampling therefore increased the risk of overfitting to an even greater extent. It is advised to use all available data for model development, and use internal validation methods such as bootstrapping or cross-validation to assess model performance. If sample size permits and multicenter data are available, then internal–external cross-validation can be considered to investigate heterogeneity in model performance across different hospitals (for example) [76].

Almost two-thirds of studies reported discrimination. Discrimination is the ability of a model to differentiate between athletes who will and who will not sustain an injury [77]. While this is an important measure of prediction performance, it is only one piece, and does not assess the reliability of these risk estimates. This is important because unreliable predictions could lead to potentially misleading clinical decisions [78]. Risk estimates may be too high for all athletes, informing clinicians that all athletes will sustain an injury, or all-risk estimates may be too low, informing clinicians that all athletes will not sustain an injury. Therefore, it is important (and widely recommended) that calibration is assessed. Calibration evaluates the accuracy of the predictions (as recommended in the TRIPOD reporting guideline) [77]. However, of the included studies, only three studies assessed calibration, with two studies using the Hosmer and Lemeshow test, which should be avoided as it provides no information on direction or magnitude of any miscalibration, and is highly influenced by sample size [78].

Only 14% of regression studies reported model equations and no machine learning studies reported tuned hyperparameters, provided a link code, or logic. Further, only 7% of eligible studies reported or adhered to the TRIPOD guidelines [25]. Complete reporting is essential to understand the research question and defined outcome, how the model was developed (including data used), and how the model performed [25]. Without these key details, it is difficult to critically appraise the methods, leading to unclear risk of bias and an inability to assess the quality of the prediction model. Further, without model transparency, through reporting equations for statistical models, or tuned hyperparameters for machine learning models, models cannot be externally validated or tested or evaluated for their outcome or economic impact. Independent and externally validated prediction models are crucial to evaluating model usefulness [79], and also reducing research waste [25]. The lack of reporting of equations, code, and/or hyperparameter tuning is consistent compared to prediction model literature in other healthcare fields, such as oncology and diabetes [24, 25, 80]. These findings suggest that poor prediction model development is inherent to the medical field, and not specific to sports medicine.

Almost half of the machine and deep learning models over- or under-sampled the outcome to control for class imbalance. Data are indicated to be imbalanced when the proportion of outcome events is unequal. Class imbalance can be an issue when calculating improper performance measures like accuracy as the larger group without the outcome will be disproportionately predicted, distorting prediction accuracy [81]. However, concerning prediction, which assesses risk, over- or under-sampling to control for class imbalance distorts the outcome prevalence, altering predictions, and thus if left unaltered (e.g., via recalibration) the model will be miscalibrated. This method will decrease prediction performance in other data sets and ultimately hinders model generalizability and application into clinical practice [30].

As with all studies there were limitations. As only three databases were included, some publications may have been missed. However, our search included over 10,000 titles and abstracts, with additional hand searching, suggesting that these overall findings would not have been altered with further study inclusion. Only studies reported in English were included, causing a single-language bias. Some studies included only machine learning models, where authors may have included different reporting and methodological parameters compared to regression models. However, there are machine learning reporting guidelines available, such as the Minimum Information About Clinical Artificial Intelligence Modelling (MI-CLAIM) checklist [82], which were not cited for any of the machine learning models, suggesting poor transparency and reporting for these studies. The TRIPOD for Machine Learning [83] (TRIPOD-AI) reporting guideline is in developmental stages, and will further aid machine learning reporting.

5 Conclusions

The majority of sports musculoskeletal injury prediction models were rated at high risk of bias and we found that, currently, there are no sports musculoskeletal injury prediction models that could be recommended for use in practice.

Typically, these models were developed with small sample sizes, and model performance was inappropriately and incompletely evaluated. Further, none of these models have been externally validated. An improvement in the methodological conduct in sports musculoskeletal injury prediction model studies is required, with an emphasis upon study analyses, with international collaborative efforts needed to acquire data of sufficient sample size for model development, external validation, and clinical impact assessment.

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Declarations

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Conflict of interest Garrett Bullock, Joseph Mylott, Tom Hughes, Kristen Nicholson, Richard Riley and Gary Collins declare that they have no conflicts of interest relevant to the content of this review.

Availability of data and material Extracted data are available through the Open Science Framework at https://osf.io/52mzn/.

Code availability Not applicable.

Author contributions GSB, TH and GSC conceived the study idea. GSB, TH and GSC were involved in design and planning. GSB and GSC wrote the first draft of the manuscript. GSB, JM, TH, KFN, RDR and GSC critically revised the manuscript. GSB, JM, TH, KFN, RDR and GSC approved the final version of the manuscript.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Patient public involvement A series of symposiums are planned at various sports medicine conferences to help further educate clinicians on this topic.

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